

CSE5104 - FL26 Project 1: Concrete Compressive Strength Analysis

This assignment can be completed individually, in pairs, or in groups of three.

Submissions are due via Gradescope on Friday, October 9 at midnight (STL time).

A one-week late extension is available with a 20% penalty applied only to the late sections, and TAs will provide rolling weekly feedback to allow for continuous revisions before the final deadline.

Part 1: Programming & Algorithm Implementation

You will analyze the UCI Concrete Compressive Strength dataset, which contains 1030 instances. Designate rows 501–630 as the testing set (130 samples) and the remaining 900 samples as the training set. Model parameters must never be updated based on the testing dataset.

<https://archive.ics.uci.edu/ml/datasets/Concrete+Compressive+Strength>

A copy is also available for download on Canvas.

Gradient Descent from Scratch:

You must implement a gradient descent optimization algorithm yourself, using Mean Squared Error (MSE) as the objective function. You may use external packages (like NumPy or scikit-learn) for matrix mathematics and feature scaling.

Take careful note of when you are expected to apply your own gradient descent algorithm, and when you should use a pre-defined tool for linear model optimization (eg. `scikit-learn.linear_model.LinearRegression`)

Model Types: Your codebase must support both univariate models ($y = mx + b$) and multivariate models (where input x is a vector of length p , resulting in $p + 1$ parameters).

Performance Evaluation: Evaluate all model fits using both MSE and variance explained

$$R^2 = 1 - \frac{MSE}{Variance(observed)}$$

Additional notes: It's highly recommended that you write code that will allow you to test different learning rates, either by accepting learning rate values as an input parameter, or by implementing an adaptive learning rate approach.

Data Analysis

Use your implemented algorithm, a third-part linear model optimization tool (eg. `scikit-learn.linear_model.LinearRegression`), and a third-party statistical tool (e.g., `statsmodels.api.OLS`) to explicitly answer the following:

Scaling & Convergence Speed:

Use your gradient descent implementation to train a multivariate model using untransformed features, and to train another using standardized or normalized features, until they both achieve a positive R^2 .

Provide a figure showing loss over iterations for both.

How does feature scaling affect convergence speed and stability? Explain why.

Scaling & The Optimal Solution:

Use an external linear model optimization tool to find the optimal solution for an untransformed multivariate model and a scaled multivariate model.

Does a linear transformation meaningfully change the optimal mathematical solution?

Non-Linear Transformations:

Apply a logarithmic transformation, $\log(x + 1)$, to your features and use your gradient descent implementation to train a new multivariate model.

Provide an updated figure showing loss over iterations.

How does this non-linear feature transformation affect convergence speed and stability?

Use an external linear model optimization tool to find the optimum solution for a logarithmically transformed multivariate model.

Construct a summary table comparing Training MSE , Testing MSE , Training R^2 , and Testing R^2 across the three optimal models: Untransformed, Standardized/Normalized, and Log-Transformed.

How does a non-linear transformation differ from a linear transformation in its impact on the optimal model solution and overall predictive performance? Explain why.

Univariate vs. Multivariate Coefficients:

Use your gradient descent implementation to train eight separate univariate models

$$y = m_i x_i + b$$

for every untransformed predictor until you achieve a positive R^2 for at least two of the eight models.

Which models were easy to train? Which models were hard?

Use an external linear model optimization tool to find the optimal solution for each of the univariate models. Compare the individual coefficients from these univariate models to the corresponding feature weights found in the optimal untransformed multivariate model.

Are they identical? Support your answer with a table of coefficients.

Why are univariate coefficients not equal to multivariate coefficients? Explain the roles of feature correlations and omitted variable bias.

Statistical Significance (p-values):

Use a statistical analysis package (e.g., statsmodels.api.OLS) to extract feature p -values.

Present a consolidated table listing all features across three configurations:

- Untransformed Multivariate Model p -values
- Standardized/Normalized Multivariate Model p -values
- Log-Transformed Multivariate Model p -values

Are there any trends between which predictors are important, and which univariate models were easy or hard to train?

Which transformation type drastically changes the statistical significance of the predictors, and which does not? Explain why.

Part 2: The Client Report

Draft a professional, non-technical data analysis report directed at a business, government, or organizational client interested in concrete strength.

Formatting & Structural Guidelines:

Word Count: 1,000 – 1,500 words (approximately 2 to 4 single-spaced pages, excluding tables and figures).

Page Formatting: Standard 1-inch margins, 11pt or 12pt clean font (e.g., Calibri, Arial, Times New Roman), and 1.15 line spacing.

Document Hierarchy: Organize using bold section headers and short paragraphs (3–5 sentences). Use bulleted lists for key takeaways to improve readability.

Figures & Tables: Every chart or table must be styled professionally, numbered sequentially (e.g., Figure 1, Table 1), accompanied by a descriptive caption, and explicitly cited in the text before it appears.

Required Sections Outline

Executive Summary

Page target: ~0.5 page

Provide a high-level summary of the analysis purpose, key findings regarding concrete mix formulations, and the primary business recommendations.

Highlight the top line performance metric (R^2) of the final model in clear, layperson terms.

Problem Context & Data Overview

Page target: ~0.5 to 1 page, excluding tables and figures

State the organization's business or engineering goals (e.g., reducing material costs, shortening curing schedules, ensuring structural safety compliance).

Explain the physical role of key concrete raw materials (Cement, Water, Blast Furnace Slag, Fly Ash, Superplasticizer, Aggregates) and curing Age.

Present a professional summary table containing descriptive statistics (mean, standard deviation, min, max) calculated on the 900-sample training dataset.

Include 1–2 figures showcasing feature distributions or key bivariate relationships (e.g., Cement or Age vs. Compressive Strength).

Model Reliability & Analytical Findings

Page target: ~0.5 to 1 page, excluding tables and figures

Explain the linear model's findings without using statistical jargon. Translate model coefficients into real-world impacts (e.g., "Holding other components constant, increasing cement by $X \text{ kg/m}^3$ yields an estimated $Y \text{ MPa}$ increase in compressive strength").

Evaluate model reliability by comparing R^2 performance on the 900-sample training set versus the 130-sample testing set. Address whether the model generalizes well to unseen mixes.

Discuss key physical interactions revealed by the data, such as the trade-off between Water-to-Cement ratios or the impact of Superplasticizers.

Strategic Recommendations & Implementation Plan

Page target: ~1 page

Present 3–4 specific, actionable operational rules based on the model results (e.g., specific batching ratios to maximize 28-day strength while controlling raw material costs).

Outline a risk mitigation plan: acknowledge model limitations (e.g., regional aggregate variations or unmodeled environmental factors).

Define a continuous monitoring framework detailing how the client should track batch quality against model predictions over time.

Part 3: Technical & Reproducibility Appendix

This section is styled as a reproducibility document for your fellow data analysts. Include your gradient descent pseudocode, hyperparameter choices (learning rate, max iterations), software dependencies, and code files/GitHub links.

Formatting & Structural Guidelines

Word Count: 1,500 – 2,200 words (approximately 4 to 6 pages, excluding code listings, embedded figures, and reference tables).

Page Formatting: Standard 1-inch margins, 11pt or 12pt clean font (e.g., Computer Modern, Calibri, Arial), single or 1.15 line spacing.

Mathematical Notation: Formulate all objective functions, gradient vectors, parameter update rules, and transformations using LaTeX or similarly clear formatting (ex. R^2 instead of R^2).

Code & Pseudocode: Present pseudocode in formatted code blocks with monospaced font. Do not dump raw Python script into the main text; link to your repository or attach full scripts in an appendix.

Required Sections Outline

Algorithmic Design & Implementation Details

Page target: ~0.5 - 1 page, excluding code blocks, tables, and figures

State the mathematical objective function.

Derive and state the gradient equation used for parameter updates.

Provide clean pseudocode for your custom gradient descent function. Detail hyperparameter settings, including learning rate α , maximum iterations, and stopping criteria.

Detail your software environment, dependencies (e.g., Python 3.x, NumPy, pandas, statsmodels), and exact dataset partition steps (900 train / 130 test split).

Feature Scaling & Gradient Descent Dynamics

Page target: ~1 page, excluding code blocks, tables, and figures

Summarize your approach and results of the Part 1 analyses:

- Scaling & Convergence Speed
- Scaling & The Optimal Solution

Include any relevant figures/tables/charts.

Linear vs. Non-Linear Feature Transformations

Page target: ~1 page, excluding code blocks, tables, and figures

Summarize your approach and results of the Part 1 analysis: Non-Linear Transformations

Include any relevant figures/tables/charts.

Univariate vs. Multivariate Model Dynamics

Page target: ~1 page, excluding code blocks, tables, and figures

Summarize your approach and results of the Part 1 analysis: Univariate vs. Multivariate Coefficients

Include any relevant figures/tables/charts.

Statistical Inference & Regression Analysis

Page target: ~1 page, excluding code blocks, tables, and figures

Summarize your approach and results of the Part 1 analysis: Statistical Significance (p-values)

Code & Reproducibility Access

Provide explicit instructions to reproduce all experimental results, including command-line arguments or notebook execution orders.

Include a working link to a public GitHub repository or specify attached code files containing your implementation.